

Week 5 – 2:

ROLL NO.:240801134

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Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Saturday, 7 December 2024, 2:08 PM
Duration	16 days 3 hours

Question **1**
Correct

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Q1) The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Note: $1 \leq N \leq 10^8$

Hint: 153 is a 3-digit number, and $153 = 1^3 + 5^3 + 3^3$.

Sample Input:

153

Sample Output:

true

Sample Input:

123

Sample Output:

false

Sample Input:

1634

Sample Output:

True

```

1  #include <stdio.h>
2  #include <math.h>
3  int main(){
4      int n;
5      scanf("%d",&n);
6      int x=0,n2=n;
7      while (n2!=0){
8          x++;
9          n2=n2/10;
10     }
11     int sum=0;
12     int n3=n,n4;
13     while (n3!=0){
14         n4=n3%10;
15         sum=sum+pow(n4,x);
16         n3=n3/10;
17     }
18     if(n==sum){
19         printf("true");}
20     else{
21         printf("false");}
22     return 0;
23 }

```

OUTPUT:

	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Q2) Take a number, reverse it and add it to the original number until the obtained number is a palindrome.

Constraints

$1 \leq \text{num} \leq 999999999$

Sample Input 1

32

Sample Output 1

55

Sample Input 2

789

Sample Output 2

66066

Code:

```
1  #include <stdio.h>
2  int main(){
3      int rn,n,nt=0,i=0;
4      scanf("%d",&n);
5      do{
6          nt=n;
7          rn=0;
8          while(n!=0){
9              rn=rn*10+n%10;
10             n=n/10;}
11         n=nt+rn;
12         i++;}
13     while(rn!=nt||i==1);
14     printf("%d",rn);
15     return 0;
16 }
```

OUTPUT:

	Input	Expected	Got	
✓	32	55	55	✓
✓	789	66066	66066	✓

Passed all tests! ✓

Q3) A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

The program should accept a number 'n' as input and display the nth lucky number as output.

Sample Input 1:

3

Sample Output 1:

33

Code:

```

1  #include <stdio.h>
2  int main(){
3      int n=1,i=0,nt,co=0,e;
4      scanf("%d",&e);
5      while(i<e){
6          nt=n;
7          while(nt!=0){
8              co=0;
9              if(nt%10!=3&&nt%10!=4){
10                 co=1;
11                 break;}
12                 nt=nt/10;}
13             if(co==0){
14                 i++;}
15                 n++;}
16                 printf("%d",--n);
17                 return 0;
18     }
19
20

```

OUTPUT:

	Input	Expected	Got	
✓	34	33344	33344	✓

Passed all tests! ✓

